Safe Steering or Near-Identity Reranking?

A Case Study of Text Detox with Embedding-Guided Candidate Selection

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Research note prepared from the Week 12 text-detox reranking experiment

https://github.com/Ily17as/Controllable-Latent-Steering-in-VAEs

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Abstract

This paper studies whether a toxicity-related signal in pretrained sentence embeddings can help select better detoxification rewrites. The pipeline embeds text with a sentence encoder, learns global and cluster-local detox directions on Civil Comments, generates prompt-based rewrite candidates with a text-to-text model, and reranks these candidates using similarity to an embedding-space target together with toxicity reduction and semantic-preservation terms. Under this formulation, the method is not a direct text decoder from steered representations; rather, it is an embedding-guided candidate-selection pipeline. The empirical picture is mixed but informative. Fixed scoring with $\alpha = 1.0$ yields substantially larger toxicity reduction than automatic per-sample selection, especially on ParaDetox, but at a noticeable semantic cost. Automatic selection is extremely conservative: the selected mean step size is nearly zero and feasible acceptance rates remain very low. Local directions are numerically stable and sometimes slightly better than global ones, but they are not the main story. The strongest conclusion is diagnostic: the current bottleneck lies less in the absence of locality itself and more in the interaction between prompt-based candidate generation, feasibility constraints, and a selection rule that frequently prefers near-identity outputs.

1 Introduction

Controllable editing methods are useful only if they change the intended attribute without destroying the underlying content. In images, this is often framed as a trade-off between target gain and identity preservation. In text, an analogous setting is detoxification: can we reduce toxicity while keeping meaning and discourse intent intact?

This note studies whether toxicity can be exploited as a human-interpretable factor in pretrained representation space. The experiment is intentionally simple. A sentence encoder produces embeddings, global and local detox directions are learned from clean-versus-toxic examples, a separate text rewriter proposes candidate paraphrases, and a reranker chooses among them. This setup isolates a practical scientific question:

Can toxicity-related directions in representation space improve candidate selection for text detox, and if not, where does the pipeline fail?

The answer is nuanced. The experiment does reveal a real detox signal: fixed scoring reduces toxicity substantially on both evaluation sets. However, the strongest finding is not a decisive win of local over global steering. Instead, the main result is that automatic per-sample step selection almost collapses to $\alpha^\star \approx 0$, while fixed scoring trades stronger detox for weaker semantic preservation. This shifts the interpretation of the study. The dominant bottleneck is not simply "we need more local directions"; it is the interaction between candidate generation, feasibility constraints, and a selection objective that often prefers near-identity outputs.

The paper makes three concrete contributions:

1. a compact benchmark comparing global versus cluster-local detox directions and fixed versus automatic scoring;
2. an empirical diagnosis showing that the current automatic policy is extremely conservative and frequently selects near-identity outputs;
3. a clarified methodological framing: the pipeline performs embedding-guided reranking of prompt-generated candidates rather than direct text generation from steered representations.

2 Problem setting and method

2.1 Embedding-guided reranking for text detox

Let $E(x) \in \mathbb{R}^d$ denote the embedding of an input text x. For each input, define an embedding-space target

$$z' = E(x) + \alpha d(x), \tag{1}$$

where $d(x)$ is either a global detox direction d_{glob} or a cluster-local direction $d_{\text{loc}}(x)$, and α is a scoring magnitude selected from a small grid.

A key implementation detail is that the system does **not** decode text directly from z', and the rewriter does not consume z' as an input representation. Instead, z' is used only at the *selection stage*: it defines a target point in embedding space against which generated candidates are scored. The text candidates themselves are produced by a separate prompt-based text-to-text rewriting model.

The global direction is learned from Civil Comments embeddings using a logistic probe and, for comparison, a clean-minus-toxic centroid difference. The main experiment uses the normalized negative probe coefficient as the detox direction,

$$d_{\text{probe}} = -\frac{w}{\|w\|_2}, \tag{2}$$

so that moving along d_{probe} shifts representations toward the cleaner side of the classifier boundary. The global probe reaches a training AUC of 0.8931 on Civil Comments embeddings.

For local scoring, the training embeddings are partitioned by MiniBatch K-Means into $K = 8$ clusters. A separate detox direction is fit inside each cluster whenever both clean and toxic classes are sufficiently represented. In the archived reranking run, all eight clusters satisfy the minimum class-count requirement, so there is no fallback to the global direction.

2.2 Candidate generation and selection

For each toxic input, the system first generates a candidate bank consisting of the original sentence and multiple prompt-based rewrites produced at prompt strengths

$$\alpha_{\text{prompt}} \in \{0, 0.5, 1.0, 1.5, 2.0\}.$$

Each candidate c is then scored using four quantities: toxicity reduction, semantic similarity to the original, a surface-form penalty, and similarity to the embedding target z'. Concretely, the archived notebook uses

$$J(c; \alpha) = w_t \, \text{target_sim}(c, \alpha) + w_d \, \Delta_{\text{tox}}(c) + w_s \, \text{sim}(c, x) - w_p \, \text{surf}(c, x), \tag{3}$$

with weights $(w_t, w_d, w_s, w_p) = (0.35, 0.35, 0.25, 0.05)$. The final run does not use a fluency term.

A candidate is declared feasible if

$$\mathrm{sim}(c, x) \geq 0.72 \quad \text{and} \quad \Delta_{\mathrm{tox}}(c) \geq 0.12. \tag{4}$$

Fixed scoring always reports the row at $\alpha = 1.0$. Automatic selection chooses the highest-objective feasible row if one exists, and otherwise the highest-objective row overall.

This distinction matters. The same candidate bank is reused across scoring values of α. Therefore, the row reported for fixed $\alpha = 1.0$ should be interpreted as the best candidate under the $\alpha = 1.0$ *scoring target*, not necessarily as a sentence generated under prompt strength 1.0.

2.3 Data and models

The detox directions are learned from a filtered Civil Comments subset. After thresholding clean and toxic examples and applying text-length filters, the archived reranking run uses 9,942 training rows. Evaluation uses 300 toxic Civil Comments examples and 150 ParaDetox examples.

The sentence encoder is `all-mpnet-base-v2`, toxicity is scored by `unitary/toxic-bert`, and the rewriter is `google/flan-t5-base`. Generation uses four return sequences and four beams.

3 Experimental protocol

The experiment compares four configurations on each evaluation set:

- global direction + automatic $\alpha^\star$,
- global direction + fixed $\alpha = 1.0$,
- local direction + automatic $\alpha^\star$,
- local direction + fixed $\alpha = 1.0$.

The main reported metrics are mean toxicity drop, mean semantic similarity, feasible acceptance rate, and mean selected α. Toxicity drop is defined as the difference between toxicity before and after rewriting, so larger is better. Semantic similarity is computed in the encoder space relative to the input. Acceptance rate is the fraction of selected outputs that satisfy the notebook's feasibility constraints.

Because the method operates through candidate reranking, the paper also reports selection diagnostics: the fraction of times the original text is selected, the fraction of selected outputs coming from the rewriter, the nontrivial-edit rate, the fraction with selected $\alpha = 0$, and the fraction for which the prompt strength matches the reported scoring α.

This is an exploratory single-run study rather than a multi-seed sweep. All quantitative conclusions below should therefore be interpreted as audited point estimates for one archived run, not as uncertainty-calibrated means.

4 Results

4.1 Main comparison

Table 1 reports the full comparison across datasets and scoring variants. Three patterns stand out immediately.

First, fixed scoring yields substantially larger toxicity reduction than automatic selection. On Civil Comments, fixed global and fixed local reduce toxicity by 0.1923 and 0.1836, whereas the corresponding automatic policies achieve 0.0927 and 0.1093. On ParaDetox, the gap is even larger: fixed global reaches 0.5139 and fixed local 0.5850, while the automatic variants reach 0.2775 and 0.3332.

Second, this stronger detox comes with lower semantic preservation. The fixed policies consistently sit lower on the similarity axis than the automatic ones. This is the expected detox-preservation trade-off.

Third, locality does not emerge as the decisive improvement. Local scoring is sometimes better than global scoring, especially on ParaDetox, but the effect is modest relative to the much larger difference between fixed and automatic selection.

Table 1: Main text detox comparison. Higher toxicity drop and higher semantic similarity are better. Acceptance rate is the fraction of selected outputs that satisfy the feasibility constraints.

Dataset	Direction	Policy	Tox. drop	Sem. sim.	Accept rate	Mean α
Civil Comments	Global	Auto	0.0927	0.9563	0.0367	0.0000
Civil Comments	Global	Fixed	0.1923	0.8912	0.0367	1.0000
Civil Comments	Local	Auto	0.1093	0.9498	0.0333	0.0017
Civil Comments	Local	Fixed	0.1836	0.9004	0.0333	1.0000
ParaDetox	Global	Auto	0.2775	0.8731	0.0133	0.0000
ParaDetox	Global	Fixed	0.5139	0.7244	0.0133	1.0000
ParaDetox	Local	Auto	0.3332	0.8512	0.0267	0.0133
ParaDetox	Local	Fixed	0.5850	0.6960	0.0267	1.0000

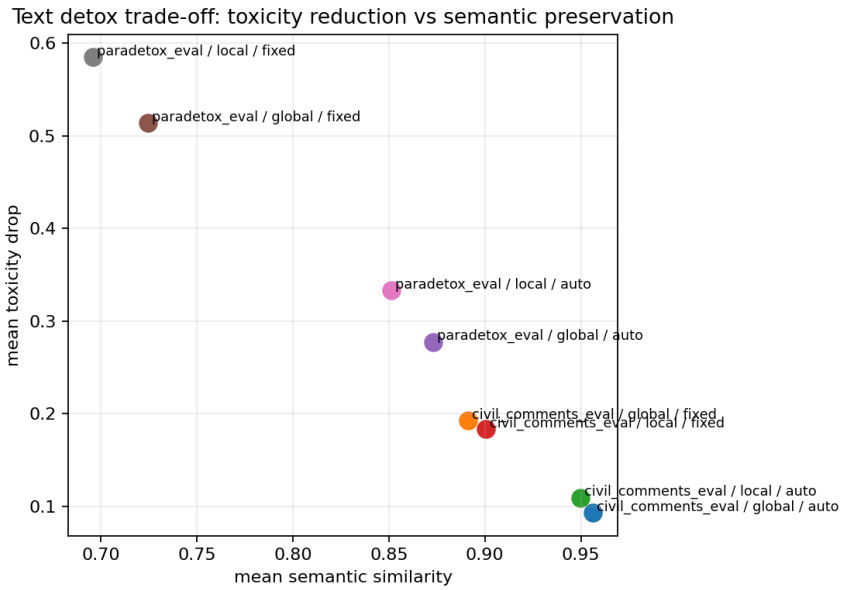

Figure 1: Text detox trade-off: toxicity reduction versus semantic preservation. Fixed scoring moves upward in toxicity reduction but leftward in semantic similarity. Automatic selection is more conservative, especially on Civil Comments.

4.2 Automatic step selection collapses toward zero

The most important diagnostic appears in the selected $\alpha^\star$ values. Global automatic selection chooses $\alpha^\star = 0$ for every evaluated example in the archived reranking run. Local automatic selection chooses $\alpha^\star = 0$ for 445 out of 450 examples and $\alpha^\star = 0.5$ only five times. The mean selected step is therefore 0.0000 for the global policy and 0.0056 over all local automatic selections, with dataset-specific means of 0.0017 on Civil Comments and 0.0133 on ParaDetox.

This behavior is visible in Figure 2 and summarized in Table 2. Automatic step selection is not discovering a broad spectrum of safe edit magnitudes. It is overwhelmingly selecting the smallest available step, which in practice means that the system often prefers either the original text or a rewrite generated from an effectively unsteered prompt.

Table 2: Automatic-policy collapse diagnostics. Rates are computed over all selected automatic outputs from both evaluation sets combined.

Mode	$P(\alpha^\star = 0)$	$P(\alpha^\star > 0)$	Rewriter sel.	Original sel.
Global	1.0000	0.0000	0.1800	0.8200
Local	0.9889	0.0111	0.2156	0.7844

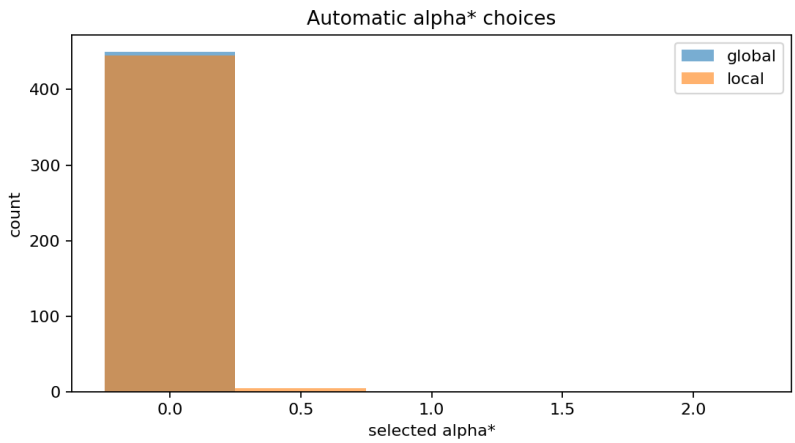

Figure 2: Automatic $\alpha^\star$ choices. The automatic policy is overwhelmingly concentrated at zero, especially for the global direction.

4.3 Selection diagnostics clarify the method behavior

The reranking view becomes especially clear in the selection diagnostics reported in Table 3. Under automatic selection, the system returns the original text in 82.0% of global cases and 78.4% of local cases. Even under fixed scoring, the original text is still selected in 65.3% of global cases and 63.3% of local cases.

Equally important, the fraction of examples for which the prompt strength matches the reported scoring α is only 0.0978 for global fixed and 0.0956 for local fixed. In other words, "fixed $\alpha = 1.0$" is usually not the same as "the final text was generated at prompt strength 1.0." It is primarily a scoring configuration over a shared candidate bank. This is exactly why the present experiment should be interpreted as embedding-guided reranking rather than direct text generation from steered representations.

Table 3: Selection diagnostics over all evaluated examples. Nontrivial edit rate coincides with the rate of selecting a non-original text.

Mode	Policy	Original sel.	Rewriter sel.	Nontrivial edit rate	Sel. $\alpha = 0$	Prompt α = sel. α
Global	Auto	0.8200	0.1800	0.1800	1.0000	0.8200
Global	Fixed	0.6533	0.3467	0.3467	0.0000	0.0978
Local	Auto	0.7844	0.2156	0.2156	0.9889	0.7889
Local	Fixed	0.6333	0.3667	0.3667	0.0000	0.0956

4.4 Acceptance rates remain very low

The feasibility filter is stringent in the current pipeline. Acceptance rates are only 3.67% and 3.33% on Civil Comments, and 1.33% and 2.67% on ParaDetox. Moreover, automatic selection does not improve this quantity over fixed scoring in the archived run. This is a useful negative result: the main issue is not just "pick better alphas"; it is that the overall candidate-generation stage rarely produces outputs that jointly satisfy detox and semantic constraints.

4.5 Local directions are stable, but not decisive

The local clustering stage itself appears numerically healthy. Figure 3 and Table 4 summarize the archived cluster statistics: cluster sizes range from 881 to 1702, the local probe AUCs lie between 0.8745 and 0.9482, and no cluster falls back to the global direction.

However, this stability does not translate into a decisive end-to-end win. Local fixed scoring is slightly worse than global fixed on Civil Comments but better on ParaDetox. Local automatic scoring is slightly better than global automatic scoring on both datasets, but the differences are small compared with the broader problem that automatic selection barely steers at all.

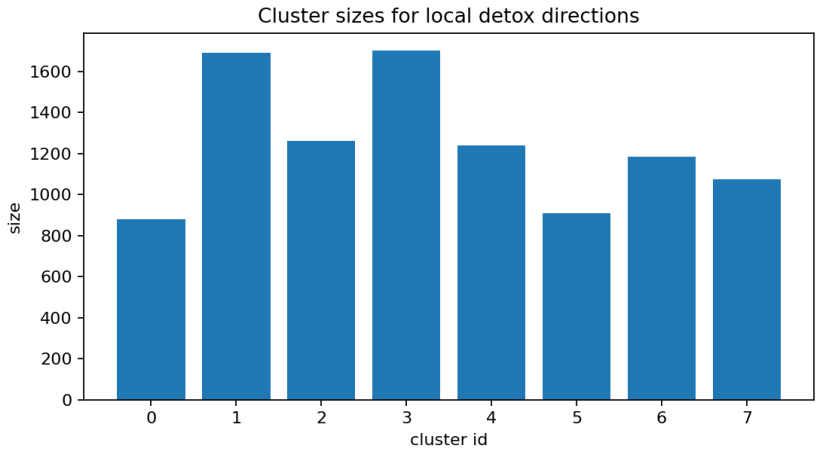

Figure 3: Cluster sizes for local detox directions. The local partition is reasonably balanced and does not show pathological tiny clusters.

Table 4: Local cluster summary for the archived reranking run.

K	Min size	Max size	Probe AUC range	Fallback rate
8	881	1702	0.8745–0.9482	0.000

5 Discussion

The strongest conclusion of this case study is not that local scoring clearly beats global scoring. The stronger and more scientifically useful conclusion is that the present automatic policy is too conservative to realize the potential of representation-guided reranking. In other words, the main bottleneck lies in the selection-and-generation loop, not only in the shape of the direction family.

Why does this happen? The current objective rewards toxicity reduction and semantic preservation simultaneously, but the feasibility thresholds and candidate quality interact in a way that often favors near-identity outputs. If most rewritten candidates are either semantically weak or stylistically unstable, the objective has a strong incentive to stay close to the original sentence. That is exactly what the selected-α histogram and the original-selection rates show.

The experiment is still useful because it isolates where the pipeline breaks:

1. the representation contains a learnable toxicity signal (probe AUC 0.8931);
2. fixed scoring can exploit that signal to reduce toxicity;
3. but automatic scoring collapses because the candidate bank and feasibility rule do not support reliable nontrivial edits.

This interpretation is more informative than a superficial "the method failed." It says the method contains a usable control signal, but the current downstream realization of that signal is weak.

5.1 Threats to validity

Three caveats matter.

First, this is a single archived run rather than a multi-seed study. The paper therefore supports diagnosis and method refinement more than final quantitative ranking.

Second, the experiment mixes three distinct components: representation scoring, a text rewriter, and an external toxicity scorer. A weak result can come from any one of these layers or from their interaction.

Third, the experiment does not implement direct text decoding from steered embeddings. It studies whether embedding-space detox directions improve candidate selection among prompt-generated rewrites. Claims should therefore be read at the reranking level, not as evidence of a full latent text decoder.

Fourth, the Civil Comments evaluation subset is defined by dataset-side toxicity thresholds, not by the toxicity scorer itself. If the scorer and dataset labels disagree, then the measured "toxicity drop" can partially reflect dataset-scorer mismatch rather than pure detox difficulty.

6 Next steps

The next iteration should tighten the study scientifically rather than only add more components. The most important next actions are:

1. **Measure nontrivial edit rate explicitly.** Report the fraction of outputs with $\alpha > 0$ and the fraction whose selected text differs meaningfully from the original.

2. **Penalize no-op solutions in the automatic policy.** For example, require a minimal detox gain before allowing $\alpha = 0$ to win, or rank feasible nontrivial candidates ahead of identity-like ones.
3. **Strengthen candidate generation.** The present rewriter frequently produces weak or degraded paraphrases. A stronger rewriting model or better prompts may improve the feasible candidate pool more than a more complicated steering rule.
4. **Align evaluation subsets with the scorer.** Build a scorer-filtered toxic evaluation split so that the detox metric is evaluated on examples the scorer itself regards as toxic.
5. **Run multi-seed comparisons.** Once the candidate-generation bottleneck is reduced, rerun the global/local and fixed/auto comparison under multiple random seeds.

7 Conclusion

This case study does not yet demonstrate a strong form of representation-conditioned text editing. Instead, it shows that toxicity-related directions in pretrained sentence embeddings can serve as useful targets for reranking prompt-generated detoxification candidates. Under this more precise interpretation, the empirical result remains informative: fixed scoring produces larger toxicity reduction than automatic selection, but at a semantic cost, while automatic selection collapses toward near-identity outputs. Local directions are stable and occasionally helpful, but they are not the dominant factor. The clearest takeaway is methodological: the present pipeline reveals a usable detox signal in representation space, but stronger text editing will require better candidate generation, better nontrivial-selection logic, and cleaner evaluation alignment.

Appendix: archived configuration and provenance

The paper is based on the archived reranking run `20260419_061121_week12_text_detox_reranking`.

The main configuration choices are:

- sentence encoder: `all-mpnet-base-v2`;
- toxicity scorer: `unitary/toxic-bert`;
- rewriter: `google/flan-t5-base`;
- local cluster count: $K = 8$;
- fixed scoring baseline: $\alpha = 1.0$;
- alpha grid: $\{0, 0.5, 1.0, 1.5, 2.0\}$;
- feasibility thresholds: semantic similarity ≥ 0.72 and toxicity drop ≥ 0.12;
- generation: four return sequences and four beams.

All scalar values in this report are transcribed from the archived CSV outputs and memo files produced by that run.